

DSA Day 34: Depth First Search & Components

Book: Data Structures Through C in Depth - Ch7 §7.12.2 · pp 364-397 | Code in Python

Overview

DFS goes deep first (recursion or stack). Used for connected components and strongly-connected components.

Key concepts to master

- recursion / explicit stack
- back edges
- connected & strongly connected components

Python implementation

```
def dfs(g, u, seen):  
    seen.add(u)  
    for v in g[u]:  
        if v not in seen: dfs(g, v, seen)  
    return seen
```

Complexity

$O(V+E)$.

Common pitfalls

- Infinite loop without visited
- Recursion depth on huge graphs

Practice

- Count connected components
- Contrast BFS vs DFS uses

